Kathan Rajendrasinh Zula

Roll No.: B23CM1063
B.Tech in Artificial Intelligence and Data Science
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Education

Degree/Certificate	Institute/Board	CGPA/Percentile	Year
B.Tech. in AI & Data Science	Indian Institute of Technology Jodhpur	8.19/10.0	2023-2027
Senior Secondary (XII)	GHSEB Board	90.94%	2022
Secondary (X)	GSEB Board	99.66%	2020

Experience

- **Undergraduate Researcher** *Jodhpur, India*
  Indian Institute of Technology, Jodhpur | Mentor: Prof. Pratik Majumdar July 1, 2026 – Present
  – Engineered a novel method to extract bias-invariant subnetworks using adversarial debiasing and masking techniques.
  – Formulated a solution to mitigate model bias entirely, eliminating the need for computationally expensive model retraining or additional datasets.
  – Implemented network pruning to heavily compress the final network size, optimizing the debiased model for highly efficient deployment on edge devices.

- **Software Development Engineer (SDE) Intern** *Remote*
  AlgoUniversity | Mentor: Manas Kumar Verma, CEO June 3, 2026 – July 11, 2026
  – Developed a custom Chrome extension to automatically extract HTML DOM trees and capture UI screenshots.
  – Integrated LLMs to accurately process the extracted data and reverse-engineer frontend Tailwind CSS code.
  – Designed multiple frontend web templates to enhance UX workflows and mentored a cohort of 30 students on technical concepts during the AI Bootcamp '26.

Projects

- **AI-Powered Policy Document Search** *Jul. 2025 – Aug. 2025*
  – **Tech Stack:** Python, TypeScript, Next.js, FastAPI, Groq LLM, FAISS, Sentence-Transformers, Vercel
  – Built a full-stack Retrieval-Augmented Generation (RAG) platform enabling semantic natural language search over policy documents, achieving sub-second queries across 1000+ chunks.
  – Engineered a modular FastAPI backend utilizing FAISS for $O(\log n)$ vector similarity search and Sentence-Transformers for local, cost-optimized embeddings.
  – Developed a responsive Next.js frontend, ensuring high answer accuracy through precise prompt engineering, LangChain semantic chunking, and source attribution.

- **Shortest Path Visualizing Project** *Sep. 2025 – Dec. 2025*
  – **Tech Stack:** Python, JavaScript, HTML/CSS, FastAPI, Docker Compose, NetworkX, OpenStreetMap
  – Architected a multi-service web application to interactively visualize map routing and shortest paths between locations using OpenStreetMap (OSM) data.
  – Designed a Python-based microservices architecture (Gateway, Map-Data, and Pathfinding) utilizing FastAPI and orchestrated with Docker Compose for seamless deployment.
  – Implemented pathfinding algorithms via NetworkX and securely fetched external geodata via HTTP clients to serve the static frontend UI.

Technical Skills

- **Languages:** Python, TypeScript, C/C++, JavaScript, SQL, Bash
- **AI/ML & RAG:** Generative AI, LLMs (Groq, GPT-3.5/4), LangChain, FAISS, Pinecone, Sentence-Transformers, NumPy
- **Web & API Dev:** Next.js, React.js, Fast API, Flask, Tailwind CSS, RESTful Microservices, Node.js
- **Databases & DevOps:** PostgreSQL, MySQL, MongoDB, SQLite3, Git, Docker, Vercel, Railway, Postman

Achievements

- **Hackathon:** Advanced to 4th elimination round at Bajaj Finserv HackerX 6.0 by building an enterprise RAG API.
- **Competitive Programming:** Peak LeetCode rating 1563; ranked 2142 in Weekly Contest 445.
- **JEE Mains (2022):** Secured 99.09 percentile among 900,000+ candidates.
- **KVPY (2021):** Qualified with an All India Rank of 4890 in the SX stream.

Relevant Coursework

- **AI/ML & Computer Vision:** Machine Learning, Deep Learning, Artificial Intelligence, Computer Vision, NLP, Optimization in ML
- **Systems & Engineering:** Autonomous Systems, Data Engineering, DSA, Design and Analysis of Algorithms
- **Mathematics & Interdisciplinary:** Probability & Statistics, Linear Algebra, Human-Machine Interaction, Data Visualization

Positions of Responsibility

- **Assistant Head,** Akshar - The Literary Arts Club, IIT Jodhpur *Aug. 2024 - Dec. 2024*
  – Coordinated event planning and managed a team of volunteers, ensuring smooth execution of literary activities and workshops.